

LAWFUL OPERATIONAL SAFEGUARDS IN AI SYSTEMS:

Surrender Recognition, Compliance Architecture, and International Humanitarian Law

Giovanni Nardacci

Founder, Human Flag Association

UNGM Partner No. 6126 · humanflag.org

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ABSTRACT

The prohibition on attacking a person recognized as hors de combat is a rule of customary international humanitarian law, reflected in Rule 47 of the ICRC Customary IHL Study and historically expressed in Article 23(c) of the Hague Regulations of 1907. It binds all parties to an armed conflict irrespective of treaty ratification. The prohibition protects the person who has surrendered or is otherwise hors de combat; its practical operation presupposes that an intention to surrender, where it arises, can be expressed, perceived, and translated into the withholding of attack before force becomes irreversible. Where a system is intended for use against persons in circumstances in which surrender is reasonably foreseeable, compliance cannot safely depend on an operational process that affords no practical means for a surrender indication to affect the engagement before that point. The protected status may arise; the operational process may nonetheless remain incapable of responding to it in time. That incapacity is not a circumstance encountered in the field. It is established in advance, by design.

For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the prohibition and contains an internal standard of recognisability: protection attaches not only to actual recognition but to what, in the circumstances, should be recognized. Article 57 supplies relevant precautionary obligations, including the duty to cancel or suspend an attack when it becomes apparent that the objective is not a military one or is subject to special protection. Article 36 provides the procedure in which the question must be examined at the stage of study, development, acquisition or adoption. For non-party States, the substantive customary obligations remain applicable; whether and through which formal procedure the corresponding system risk is reviewed depends on the State's national law, weapons-review policy, procurement framework or military directives. Customary precautionary duties remain relevant to the substantive assessment but are not treated here as creating a universal Article 36-equivalent procedure.

This paper argues that surrender-recognition capability, understood as the preservation of a recognition window, is not a new obligation but a condition of effectiveness of obligations that have bound parties for over a century. The recognition window is an analytical and testable measure of whether the operational process affords sufficient

time for a surrender indication, where one arises, to affect the engagement before force becomes irreversible; the paper does not treat that interval as an independently stated rule of customary law. It advances a determinate proposition: where the foreseeable normal use of a system would engage persons who, in the circumstances, should be recognized as hors de combat, review must examine whether the system preserves the practical possibility of compliance; and the systematic exclusion of technically achievable safeguards relevant to that question warrants documented justification rather than silence.

I. INTRODUCTION

Debates concerning artificial intelligence safeguards in operational and military-adjacent environments are often framed in binary terms: more safeguards reduce operational effectiveness, fewer safeguards increase capability. This framing is incomplete. International Humanitarian Law imposes obligations concerning the treatment of persons hors de combat, including individuals who clearly express surrender or otherwise cease participation in hostilities. These obligations predate modern AI systems by decades and remain legally operative independent of technological evolution.

For purposes of this paper, 'military-adjacent' systems are systems that do not themselves release force but materially influence targeting, escalation, prioritization, or engagement decisions — for example, decision-support, battlefield intelligence, and targeting-recommendation architectures. The term does not extend to logistics, general-purpose intelligence analysis, or communications infrastructure without a targeting- or escalation-relevant function. For the purposes of the legal review analysis in Section III, such systems are relevant where they form part of, materially enable, or determine the operation of a weapon, means, or method of warfare — not by virtue of their military use alone.

The emergence of increasingly autonomous decision-support systems, targeting architectures, battlefield intelligence systems, and military-adjacent AI raises a distinct question: whether certain safeguards should be understood not merely as optional policy restrictions, but as operational compliance mechanisms relevant to existing humanitarian obligations. Existing scholarship has examined extensively whether autonomous systems may comply with IHL obligations as a general matter, and has addressed meaningful human control as a possible compliance criterion.

It has not, however, yet fully isolated surrender-recognition safeguards as a distinct compliance-relevant capability class, separate from the broader meaningful-human-control debate. This paper addresses that gap. It focuses on surrender recognition capability in operational AI systems as an illustrative compliance-relevant safeguard class. It does not address consumer AI, general-purpose language models, or systems without operational or escalation-relevant functions.

The paper advances no claim that international law mandates any specific implementation architecture. Its claim is structural. The source of the obligation is customary international humanitarian law, reflected in Rule 47 and Article 23(c) of the Hague Regulations of 1907. That prohibition binds all parties to an armed conflict irrespective of treaty ratification. For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the same prohibition; Article 57 supplies precautionary support; and Article 36 provides the procedural anchor for review at procurement. The question this paper poses is what those obligations require when an autonomous function compresses the engagement below the recognition window — the interval in which an intention to surrender can be expressed, detected, assessed and translated into the withholding of attack before force becomes irreversible. The argument is doctrinal-analytical; its qualifications are consolidated in the Methodological Note rather than repeated throughout the text.

II. EXISTING OBLIGATIONS UNDER INTERNATIONAL HUMANITARIAN LAW

A. Customary foundation: Rule 47 and Article 23(c) of the Hague Regulations

The starting point is a rule no delegation disputes. Attacking a person who is recognized as hors de combat is prohibited under customary international humanitarian law, reflected in Rule 47 of the ICRC Study on Customary International Humanitarian Law and applicable in both international and non-international armed conflict. The prohibition's historical root is Article 23(c) of the Hague Regulations of 1907, which forbids killing or wounding an enemy who, having laid down arms or no longer possessing means of defence, has surrendered at discretion. The Hague Regulations are widely accepted as declaratory of customary law, a proposition affirmed by the International Military Tribunal at Nuremberg and by the International Court of Justice. The prohibition therefore long predates Additional Protocol I and does not originate in it.

The prohibition binds the party employing the means or method of warfare. Delegating detection, classification, selection or engagement functions to an autonomous system does not transfer the obligation to the machine, nor does the machine's inability to process surrender make the obligation legally inoperative. The relevant question remains whether the party selected a system and conditions of use capable of respecting the prohibition in reasonably foreseeable circumstances.

The prohibition operates upon recognition. It is engaged when a person is recognized to be hors de combat; and recognition depends on a practical sequence: an intention must be expressed, it must be perceived, it must be assessed, and it must be translated into the withholding of attack before force becomes irreversible. Each step consumes time. The sequence therefore has a minimum duration below which it cannot be completed at all. Where autonomous functions compress an engagement below that minimum duration, the difficulty is not that a surrender signal was given and missed. It is that no

meaningful opportunity to give one existed, and that no recognition could therefore occur. The prohibition remains applicable, but the operational process may prevent the legally relevant facts from affecting conduct before the point of irreversibility.

This paper refers to that minimum interval as the recognition window. The window is not proposed as new law, and it is not treated as an independently stated rule of customary law. It is an analytical and testable measure of whether the operational process affords sufficient time for a surrender indication, where one arises, to affect the engagement before force becomes irreversible. For States party to Additional Protocol I, the measure is particularly relevant to the express "recognized or should be recognized" standard in Article 41; for all parties, it operationalizes the question whether the prohibition can in practice affect conduct. It is also measurable: the end-to-end interval a system affords, under stated conditions, can be determined, tested and recorded.

Article 23(c) establishes the prohibited result. The recognition window does not add a new rule; it tests whether the operational process permits that rule to affect conduct in time. Where a system is intended for use against persons and surrender is reasonably foreseeable before force becomes irreversible, the employing party must assess whether the complete operational arrangement can respond to that change of status; where it cannot, the relevant response may be additional safeguards, effective human intervention, or restrictions on the circumstances of use.

B. Common Article 3

Common Article 3 of the Geneva Conventions establishes minimum protections applicable during armed conflict. Its text is direct: "Persons taking no active part in the hostilities, including members of armed forces who have laid down their arms and those placed hors de combat by sickness, wounds, detention, or any other cause, shall in all circumstances be treated humanely." The obligation does not depend upon advanced technological conditions. It applies independent of weapon type or operational medium; the ICRC Commentary confirms that the provision establishes a minimum humanitarian floor applicable across all forms of armed conflict regardless of the means employed.

As operational systems become increasingly dependent on automated identification, autonomous classification, machine-assisted targeting, and algorithmic escalation processes, the practical implementation environment changes. The underlying obligation does not. The legal question is whether system architectures that systematically fail to process or preserve surrender-relevant indicators create heightened risks of non-compliance with obligations that have been operative for more than seven decades.

C. Additional Protocol I, Article 41: treaty formulation and recognisability standard

For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of the customary prohibition examined above. Article 41(1) provides: "A person who is recognized or who, in the circumstances, should be recognized to be hors de combat shall not be made the object of attack." Article 41(2) defines the protected condition: a person is hors de combat if he is in the power of an adverse Party, "clearly expresses an intention to surrender", or is incapacitated, provided he abstains from any hostile act and does not attempt to escape.

The wording of paragraph 1 deserves attention. Protection attaches not only to surrender that is in fact recognized, but also to surrender that, in the circumstances, should be recognized. The provision therefore contains an internal standard of recognisability. It does not itself prescribe a sensor, classifier, signalling protocol or control architecture. Nor does it impose a positive obligation to capture, hold or evacuate. Its relevance is structural: where engagement processes are mediated by automated classification, the circumstances in which recognition should occur are partly determined by the architecture selected. An architecture that affords no recognition window does not lower the standard; it may foreclose the practical conditions under which the standard can be met.

A textual boundary should be stated. Article 42 of Additional Protocol I expressly requires that a person parachuting from an aircraft in distress "shall be given an opportunity to surrender" before being made the object of attack. Article 41 contains no comparable express language. This paper therefore does not claim that Article 41 imposes a positive obligation to create an opportunity to surrender. It claims only that the recognisability standard internal to Article 41 presupposes an interval, and that a review may examine whether a system preserves that interval. The distinction is material: the argument is not that the prohibition contains the recognition window as part of its content; it is that the recognition window is a precondition of the prohibition's practical operability.

Article 41 operationalizes recognition through identifiable conditions and observable indicators. The ICRC Commentary explicitly notes that raised arms, abandonment of weapons, and verbal surrender expressions may qualify as indicators regardless of the medium through which they are communicated.

The question is therefore not whether AI systems become "subjects" of humanitarian law. The question is whether deployment architectures that systematically omit surrender-relevant recognition capabilities undermine the ability of human operators or operational chains to satisfy a recognition duty the treaty states in terms.

D. Additional Protocol I, Article 57: precautionary support

Article 41 is the direct recognition rule. Article 57 supports it procedurally. Article 57(2)(a)(i) requires those who plan or decide upon an attack to "do everything feasible

to verify that the objectives to be attacked are neither civilians nor civilian objects and are not subject to special protection" — a provision directed in the first instance at the identification of military objectives. Its relevance to surrender is functional rather than direct: feasible verification and feasible precautions require that status-changing information capable of making a person no longer targetable be available to the decision-maker, whatever the source of that information. Where the decision is reached through, or on the recommendation of, an automated system, the feasibility of verification is partly an architectural property: a system that cannot represent surrender-relevant states cannot supply them to the verifier.

The precautionary framework also governs the engagement as it unfolds. Article 57(2)(b) requires that an attack be cancelled or suspended if it becomes apparent that the objective is not a military objective or is subject to special protection; a corresponding obligation, qualified by feasibility, is recorded as customary in Rule 19 of the ICRC Study. That duty presupposes that where legally relevant information, including a change of status arising during the engagement, becomes available before the point of irreversibility, the architecture has not already made cancellation or suspension impossible by design. The pairing of Article 57(2)(a) and (2)(b) therefore covers both moments: the initial choice and verification of means, and the response to new information during the engagement.

This reading is not confined to States party to the Protocol. The duty to do everything feasible to verify that targets are military objectives is customary (Rule 16 of the ICRC Study), as is the prohibition on attacking persons hors de combat (Rule 47) and the prohibition of denial of quarter (Rule 46; AP I Article 40), binding on all parties to a conflict regardless of treaty ratification status.

An anticipated objection should be addressed here, because this is where it arises. It may be said that IHL imposes behavioural duties on human beings — to recognize, to verify, to refrain — and not technical capabilities on machines. The premise is correct; the conclusion does not follow, for two textual reasons. First, the precautionary regime itself extends beyond conduct in the moment of engagement: Article 57(2)(a)(ii) requires "all feasible precautions in the choice of means and methods of attack". The selection of the means is itself the object of a duty — and an engagement architecture is a means. Second, the recognition standard of Article 41(1) is indexed to "the circumstances", and the party that selects a machine-speed engagement process selects the circumstances in which recognition must occur. A party cannot rely on the impossibility of discharging a duty when that impossibility is the product of its own choice of means: the behavioural duty remains human, but its feasibility is determined upstream, at the level of design and procurement — which is precisely where legal review places the obligatory examination, for States bound by such procedures. The argument therefore requires no transfer of legal personality to the machine; it requires only that duties about the choice of means be taken to mean what they say.

The convergence extends to the most significant non-party. The United States Department of Defense, in its July 2023 update to the Law of War Manual, added a dedicated section on "Feasible Precautions to Verify Whether Objects of Attack Are Military Objectives" and revised its guidance to recognise a presumption of protected status absent information indicating a military objective. The verification duty, in other words, is stated in terms on both sides of the treaty boundary.

E. Command Responsibility

The analysis of lawful operational safeguards intersects with the doctrine of command responsibility under Article 87 of Additional Protocol I and Article 28 of the Rome Statute. These provisions establish that commanders may incur individual responsibility for crimes committed by forces under their effective control where the commander knew or should have known of the risk and failed to take necessary and reasonable measures to prevent or repress such acts. Whether the absence of a surrender-recognition safeguard becomes relevant to a command-responsibility analysis depends on the ordinary elements of that doctrine — knowledge or constructive knowledge of the risk, foreseeability of the specific harm, effective control over the forces and the procurement decision, the existence of operational alternatives that were not adopted, and a causal link between the omission and the outcome — rather than following automatically from the absence of the safeguard alone. As the temporal analysis in Section IV makes explicit, where the engagement window excludes meaningful human intervention, the architectural decision taken at procurement — rather than the operator's split-second choice — becomes the locus at which these elements, including effective control and foreseeability, are most plausibly assessed.

III. THE LEGAL REVIEW OBLIGATION

Every deploying State assesses the legality and safety of new weapons before fielding them. The name of the procedure, its legal basis, and the authority at which residual risk is accepted differ from State to State. For States party to Additional Protocol I, the procedure is codified in Article 36. For the United States, it is conducted under DoD Directive 3000.09 and the Law of War Manual, supported by the system-safety framework of MIL-STD-882E. For other States, the substantive customary obligations remain applicable, and the corresponding system risk is reviewed, where it is reviewed at all, under national law, weapons-review policy, procurement frameworks or military directives. Customary precautionary duties bear on the substantive assessment; they are not treated here as creating a universal Article 36-equivalent procedure.

The obligation whose performance is at stake here is customary and binds every deploying State. The procedure in which the question arises differs in name, but the question itself does not change: whether the system's foreseeable normal use complies with the rules of armed conflict, including the prohibition on attacking persons hors de

combat and the rule of distinction. Where a person may become hors de combat during the engagement cycle, a review that has not examined whether the system preserves the practical possibility of recognition has not completed its determination, whatever the procedure is called.

Article 36, for States party to Additional Protocol I, provides: "In the study, development, acquisition or adoption of a new weapon, means or method of warfare, a High Contracting Party is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited by this Protocol or by any other rule of international law applicable to the High Contracting Party."

Three features of the Article 36 obligation carry the argument. First, its temporal scope: the review attaches to "study, development, acquisition or adoption" — the ICRC Guide confirms that this covers all stages of the procurement process, from conception onward. Article 36 therefore operates at exactly the point where architectural choices, including the inclusion or exclusion of recognition capabilities, are made and are still reversible. Second, its material scope: the review must assess compatibility not only with the Protocol's specific weapons prohibitions but with "any other rule of international law applicable" — which includes the recognition duty of Article 41(1) and the verification duty of Article 57(2)(a)(i) examined above. Third, its mandatory character: for States party to the Protocol, Article 36 is not a recommendation; it is "an obligation".

The consequence can be stated plainly. Whether an AI-enabled operational system is capable of identifying, preserving, or escalating surrender-relevant indicators within its engagement window is not a question this paper invents and offers for consideration. It is a question that legal review — under whichever procedure applies — is already required to ask, because the employment of a system incapable of supporting recognition and verification "in some or all circumstances" is precisely what the review exists to determine. The contribution of this paper is to make the question explicit and to supply criteria — set out in Sections V and VII — by which it can be answered in a documented, auditable form, whichever procedural route applies.

Nor can the novelty of the technology displace the inquiry. The International Court of Justice, in its 1996 Advisory Opinion, held that the established principles and rules of humanitarian law apply "to all forms of warfare and to all kinds of weapons, those of the past, those of the present and those of the future." The temporal openness of the law is judicially settled; legal review, where it applies, is its procedural expression.

IV. THE TEMPORAL FOUNDATION: HUMAN DECISION-MAKING AND MACHINE-SPEED OPERATIONS

The analysis developed in this paper rests upon a premise that is often acknowledged but insufficiently examined: the problem of time. International Humanitarian Law presumes the existence of decision-makers capable of assessing circumstances,

interpreting observable conduct, and applying legal obligations before force is employed. Historically, these functions have been performed by human actors operating within biological and cognitive constraints.

A. Human Decision-Making Within Biological Limits

Human cognition operates within measurable temporal limits. Experimental research places basic perceptual processing on the order of one hundred milliseconds, with motor response adding a further fifty to one hundred milliseconds; simple reaction times generally fall between roughly 150 and 300 milliseconds. These figures, however, describe simple reaction — a single pre-cued response to an expected stimulus. The decision relevant to the lawful use of force is categorically different. It requires perception, recognition, discrimination between combatant and hors de combat status, evaluation of alternatives, and a judgment to act or to refrain. Such choice-and-judgment responses extend substantially beyond the simple-reaction range and increase with task complexity.

This distinction is particularly significant in military and security environments, and the operational context tends to degrade rather than improve performance. Acute stress, cognitive load, fatigue, and environmental ambiguity have measurable effects on both reaction time and decision quality. Experimental work on lethal-force decision-making confirms that engagement and restraint decisions cannot be reduced to reflexive reactions: they are measurable in the hundreds-of-milliseconds-to-seconds range and are sensitive to stressor conditions. The human decision to use, or to withhold, lethal force is therefore not a sub-second act; it is a deliberative process whose realistic floor lies well above the simple-reaction range and whose practical duration, under operational stress, may extend to seconds.

B. Machine-Speed Operations

Autonomous and AI-assisted systems operate within a fundamentally different temporal domain. Computational processes associated with detection, classification, prioritization, targeting support, and engagement recommendation may occur within milliseconds or seconds, depending upon system architecture and operational design. Defensive systems such as Phalanx and SeaRAM illustrate the compression: publicly available technical descriptions indicate that Phalanx-class close-in weapon systems may confirm a threat roughly thirty seconds before impact and reach effective engagement range only some twelve seconds before impact, leaving on the order of eighteen seconds of engagement window once activated — a sequence that, once triggered, leaves little practical room for human intervention.

More generally, a system that classifies an object as a potential threat may initiate an engagement within milliseconds — long before a human operator has the opportunity to

assess the situation or intervene effectively. As autonomy increases, these cycles contract further; the UNIDIR analysis of the interpretation and application of IHL to lethal autonomous weapon systems observes that calculations performed at machine speed may eclipse any horizon for timely, meaningful human intervention.

C. Temporal Asymmetry and the Compression of Human Control

The significance of this disparity does not depend upon any single numerical threshold. What remains constant is the existence of a structural temporal asymmetry between biological cognition and computational processing: the machine's decision-to-action cycle may operate within a window — milliseconds to a few seconds — that falls below the threshold required for meaningful human assessment of the engagement. This phenomenon has increasingly been described as temporal compression. As operational systems become faster, the time available for human intervention contracts; decision windows that once permitted independent human assessment may shrink to the point where human participation becomes largely procedural. Under such conditions, the mere presence of a human operator within the operational chain does not establish meaningful human control: where machine-generated outputs are produced faster, and in greater volume, than they can realistically be assessed, operators become dependent upon automated recommendations, with the documented risks of automation bias and "rubber-stamping". A human may remain technically "in the loop" while the loop itself runs faster than human cognition.

The implications for International Humanitarian Law follow directly. If the practical opportunity for independent human evaluation diminishes as system speed increases, compliance with legal obligations cannot rely exclusively upon external human intervention; compliance-relevant functions must be preserved within the system architecture itself. The argument advanced in this paper proceeds from this observation: surrender recognition capability is examined not as an ethical preference, but as a lawful operational safeguard capable of preserving compliance-relevant information within decision environments characterized by temporal compression. The relevant question is not whether autonomous systems should be faster or slower than human operators, but whether operational architectures preserve sufficient opportunity for compliance with existing legal obligations when machine decision cycles outpace meaningful human assessment.

The structure of this argument — a duty whose exercise presupposes a humanly bridgeable time window, a class of systems that closes the window, and the consequent migration of the duty into design — is not unique to humanitarian law.

V. SURRENDER RECOGNITION AS A CAPABILITY CLASS

A. Definition and Technical Feasibility Criteria

For purposes of this paper, surrender-recognition capability is understood as the technical capacity of an operational system to preserve the recognition window — the minimum interval required for an intention to surrender to be expressed, detected, assessed and translated into the withholding of attack before force becomes irreversible. It encompasses technical mechanisms designed to: identify surrender expressions; recognize de-escalation indicators; preserve uncertainty during ambiguous encounters; inhibit automatic escalation under uncertainty; escalate ambiguous cases to human review; and maintain operational awareness of hors de combat conditions within that interval.

The concept does not require perfect recognition. Human operators themselves routinely operate under uncertainty; IHL does not demand infallibility. The relevant inquiry concerns whether foreseeable and technically achievable mitigation mechanisms are systematically excluded. The term "technically achievable" is not an abstract standard. For purposes of procurement and audit analysis, feasibility should be assessed against documented maturity criteria, such as: Technology Readiness Level (TRL) of relevant sensor and classification components; acceptable inference latency thresholds for the operational tempo of the system in question; availability of annotated training datasets for surrender-relevant signal categories in representative environments; and capacity for independent validation of recognition performance against defined recall and precision benchmarks. Established frameworks such as the NIST AI Risk Management Framework and the IEEE 7000 series provide further reference criteria for structured feasibility assessments. These criteria are offered as anchors for a documented assessment, not as thresholds this paper purports to fix: where the maturity line should sit for a given system, or what recall is adequate for a given tempo, are determinations for the reviewing authority — the legal point is that the determination must be made, documented, and reviewable, rather than left implicit in a procurement silence.

The latency criterion is the direct operational expression of the temporal analysis in Section IV: for a safeguard to be compliance-relevant, its inference latency must fall within the decision window of the system in which it operates. A surrender-recognition mechanism that operates at machine speed addresses precisely the gap that human oversight, under temporal compression, cannot fill. This framing draws on the principle of precaution in attack reflected in Article 57: where surrender recognition constitutes a technically achievable precautionary mechanism against the criteria above, its systematic exclusion requires justification within the feasible-precaution analysis — and, where applicable, within legal review.

B. Technical Forms

Potential implementation forms include: multimodal surrender signal recognition; gesture recognition; linguistic surrender detection; uncertainty escalation thresholds;

human override escalation pathways; refusal-to-engage constraints under unresolved surrender ambiguity; and probabilistic de-escalation weighting. This paper intentionally avoids endorsing any single implementation architecture. The legal relevance arises not from a particular model design, but from the operational function served.

C. Operational Relevance

In operational environments, escalation errors occur not only through aggression, but through the absence of de-escalation mechanisms. A system optimized exclusively for hostile detection without surrender-sensitive mitigation pathways increases: false hostile continuation; escalation persistence; failure to preserve surrender ambiguity; and downstream attribution risk. Under such circumstances, the absence of safeguards does not increase operational safety or national security; in some environments, it foreseeably reduces both.

D. Illustrative Operational Scenarios

The following scenarios are illustrative only, intended to concretize the compliance risk analysis rather than to assert factual conclusions about any specific system or deployment.

Scenario 1 — ISR drone in dense urban environment. An ISR drone operating in a densely populated urban environment is equipped with an automated threat classification system, optimized to identify hostile actors and flag targets for human-authorized engagement. It includes no mechanism for detecting or flagging surrender-relevant behaviors — raised hands, discarded weapons, stationary posture following prior movement — as a distinct classification state. In an engagement sequence, a combatant who has ceased hostile activity and raised hands is classified as "still present in threat zone". The system provides no signal to the human operator that a potentially hors de combat condition may exist. The operator, relying on the automated classification, proceeds with engagement authorization. The architecture has foreclosed a compliance-relevant decision pathway before the human operator was engaged — and, in the language of Article 41(1), has shaped the "circumstances" in which surrender should have been recognized. Whether a legally relevant failure occurred depends on facts not assumed here; what the scenario establishes is the locus of the problem: design, not operator judgment.

Scenario 2 — Decision support system for indirect fire. A decision support system for indirect fire operations processes sensor fusion data to provide targeting recommendations to a fire mission authority, ranking potential targets by threat probability. It includes no uncertainty escalation logic for scenarios in which available data is insufficient to distinguish active hostile engagement from cessation of hostilities or withdrawal. A fire mission authority, operating under time pressure that reflects the

temporal compression identified in Section IV, authorizes engagement without independent assessment of hors de combat status. Post-engagement review reveals that sensor data available at the time of the recommendation contained indicators consistent with cessation of hostile activity that the system's architecture was not designed to process or flag. The absence of uncertainty escalation logic has reduced the practical operability of the verification duty under Article 57(2)(a)(i); the procurement decision not to include such logic is exactly the kind of determination that legal review — where applicable — exists to examine.

VI. LAWFUL OPERATIONAL SAFEGUARDS

A. Distinguishing Policy Safeguards from Compliance-Relevant Safeguards

Public and regulatory discussions frequently treat all AI safeguards as belonging to a single undifferentiated category. This obscures an important distinction. Some safeguards are discretionary policy constraints reflecting corporate preference, reputational concern, or product strategy. Others perform functions directly relevant to legal compliance. The latter category can now be defined: a lawful operational safeguard is a technically implementable system function whose absence foreseeably degrades compliance with an existing legal obligation within the decision window in which operational outcomes are determined. Each element of the definition does work: "technically implementable" excludes the aspirational; "foreseeably degrades" anchors the category in established risk analysis rather than hindsight; "existing legal obligation" excludes preference and policy; and "within the decision window" incorporates the temporal criterion of Section IV, distinguishing safeguards that operate where the decision is actually made from oversight that arrives after it. Such safeguards are not mandated in explicit technical form by existing treaties; they are the foreseeable mitigation mechanisms through which preexisting legal obligations — recognition, verification, precaution — remain capable of being discharged in machine-mediated decision environments.

B. Compliance Architecture

Modern operational systems increasingly depend upon algorithmic classification, probabilistic targeting, autonomous assistance, machine-mediated escalation, and automated operational filtering. As a result, legal compliance increasingly interacts with system architecture. This interaction is reinforced by temporal asymmetry: as operational decision cycles accelerate, legal compliance cannot rely exclusively on post hoc human intervention; certain compliance-relevant functions must be preserved within the architecture itself, because the available decision window is insufficient for meaningful human review. Under this framework, such safeguards function not as "ethical add-ons" but as compliance-preserving mechanisms, operational risk

mitigations, uncertainty preservation structures, and escalation-limiting controls. Their legal significance arises not from moral aspiration, but from operational consequence.

C. Removal as Legal Risk

Where technically achievable safeguards relevant to surrender recognition are intentionally excluded from operational AI systems, foreseeable risks arise: increased escalation under uncertainty; increased probability of failure to recognize hors de combat conditions; reduced human review opportunities; and weakened operational restraint pathways. Removal of such safeguards is therefore not a neutral operational choice; it is a design determination that engages the duties examined in Sections II and III and warrants documented justification. This does not imply automatic illegality. It means that safeguard absence is relevant to legal and operational evaluation — including command responsibility analysis, as discussed in Section II.E, and, where applicable, legal review, as discussed in Section III.

For audit and procurement purposes, systematic exclusion may be evidenced by one or more of the following indicators: procurement specifications that explicitly exclude surrender-relevant signal classes from system requirements; absence of hors de combat or de-escalation scenarios in pre-deployment validation protocols; technical documentation or supplier statements indicating that surrender recognition is not a supported or intended capability; or architectural design choices that remove or preclude human escalation pathways in ambiguity-resolving contexts.

VII. TECHNICAL LIMITATIONS AND BOUNDARY CONDITIONS

Any credible analysis of surrender recognition as a compliance-relevant capability must acknowledge significant technical limitations. These limitations do not defeat the argument; they define its proper scope.

A. False Positives, Adversarial Signaling, and Robustness

Recognition systems may generate false positives, incorrectly classifying non-surrender gestures or communications as surrender indicators. In adversarial environments, actors may deliberately exploit surrender recognition mechanisms to create operational pauses, gain tactical advantage, or manipulate escalation dynamics. Research on adversarial examples in deep learning demonstrates that small, deliberately crafted perturbations can cause classification systems to produce incorrect outputs with high confidence; gesture spoofing, acoustic mimicry, and signal injection are real vulnerabilities and must be addressed through structured red-teaming, adversarial scenario testing, and robustness validation as part of any pre-deployment audit process.

However, the existence of adversarial risk does not resolve the underlying legal question. IHL itself operates under conditions of deception — it prohibits perfidy (Article 37) precisely because false surrender is a known tactic — and it does not suspend obligations because of the risk of false signals. The relevant question is whether the system architecture provides any meaningful pathway for surrender-relevant information to influence operational outcomes, and whether the system has been tested against adversarial conditions as part of a structured assurance process.

Recognition systems may also generate false negatives — failing to identify genuine surrender indicators due to sensor limitations, model underfitting, or threshold calibration optimized for precision over recall. In IHL terms, a false negative directly implicates the verification duty of Article 57(2)(a)(i), as it increases the risk of attacking persons hors de combat. Where operational systems are calibrated to minimize false positives at the expense of recall, the resulting degradation in surrender-signal detection requires specific justification under feasible precaution analysis. This does not imply that perfect recall is required; it means that the calibration rationale should be documented and reviewed as part of compliance-relevant assurance processes.

B. Battlefield Ambiguity and Imperfect Recognition Environments

Operational environments present conditions that may substantially degrade recognition performance: low visibility, sensor degradation, partial data, communication interference, and high-tempo decision chains. These conditions may constrain the practical implementation of surrender recognition mechanisms in certain operational contexts. They do not eliminate the analytical relevance of the capability class. The principle of feasible precaution under Article 57 is explicitly conditioned on what is achievable under the circumstances; where recognition capability is not feasible given the maturity and operational parameters of the system, the analysis changes accordingly. Even where uncertainty cannot be eliminated, the temporal asymmetry of Section IV remains relevant: the challenge is not only recognition accuracy but the preservation of sufficient decision space for lawful assessment. The paper does not claim that surrender recognition capability is universally achievable. It claims that its systematic exclusion from systems where it is technically feasible — assessed against the criteria in Section V.A — engages the duties of recognition, verification, and review.

C. Scope Limitation: Operational Systems Only

The analysis in this paper is confined to AI systems with operational, targeting, escalation, or battlefield-relevant functions, as defined in Section I. It does not apply to consumer AI, general-purpose language models, logistics systems, or administrative applications. This scope limitation is analytically significant: the compliance-relevance of

surrender recognition capability arises specifically from the operational consequence of its absence in systems capable of influencing kinetic or escalatory outcomes.

VIII. IMPLICATIONS FOR FUTURE GOVERNANCE

A. Autonomous Systems and International Governance

As AI systems become increasingly integrated into operational, military-adjacent, intelligence, and escalation-sensitive environments, the distinction between discretionary alignment restrictions and lawful operational safeguards will become increasingly relevant in autonomous systems governance, dual-use AI regulation, military AI procurement, operational accountability analysis, and IHL adaptation debates — including the GGE LAWS process.

B. European Regulatory Context and Dual-Use Governance

The EU Artificial Intelligence Act expressly excludes AI systems developed or used exclusively for military purposes from its scope under Article 2(3); the framework proposed here makes no claim of direct applicability of the Act to military AI systems. The European regulatory landscape remains relevant through two channels. First, dual-use AI systems may fall within the Act's high-risk classification under Annex III, in which case its requirements for risk management, technical documentation, and human oversight intersect with the safeguard analysis proposed here. Second, European and NATO member state procurement frameworks are subject to national standards, dual-use export controls, and interoperability requirements that create practical compliance expectations and procurement constraints; the concept of foreseeable misuse as a risk analysis criterion engages compliance-relevant questions under both IHL and European risk governance frameworks even where the AI Act does not directly apply.

C. Procurement Compliance Framework

The procurement framework proposed in this section maps onto existing legal and institutional review processes — first and foremost legal review as examined in Section III, where applicable, which provides its natural procedural home. NATO AI assurance initiatives and national frameworks such as the UK MoD's AI approach and US DoD Directive 3000.09 increasingly emphasize structured testing, documentation, and human oversight criteria that align with the checklist elements below. The following framework is offered as an operational checklist for acquisition authorities, certification bodies, and independent auditors evaluating operational AI systems with escalation-relevant functions.

1. Technical Feasibility Assessment. Has surrender recognition capability been assessed as technically feasible for this system in its intended operational environment? Has the assessment been documented with reference to maturity criteria, sensor specifications, and latency constraints? Has it been conducted or reviewed independently of the system developer?

2. Exclusion Justification. Where surrender recognition capability has been excluded, has the exclusion been documented with specific operational justification? Has the justification addressed whether the exclusion is permanent or context-specific? Has the foreseeable IHL compliance risk created by the exclusion been documented and reviewed?

3. Human Escalation Pathways. Does the system architecture include a "pause for verification" pathway triggered by surrender-ambiguous indicators? Is human escalation available before irreversible action in unresolved de-escalation scenarios, and is it feasible within the system's decision window (Section IV)? Are escalation pathways documented in operational procedures and tested under realistic conditions?

4. Pre-Deployment Audit. Has an independent pre-deployment audit been conducted using surrender and de-escalation scenarios? Have adversarial signaling and spoofing scenarios been included in the audit test suite? Has audit documentation been retained for post-deployment operational review?

5. Command Responsibility Documentation. Has authorizing command reviewed and acknowledged the IHL compliance risk profile of the system? Is there a documented record of the feasible precaution analysis — and, where applicable, of legal review — conducted prior to deployment authorization?

IX. LIMITATIONS OF THE ARGUMENT

This paper does not claim: that current international law prohibits AI systems; that all autonomous systems are unlawful; that surrender recognition is technologically solved; that any single safeguard architecture is legally mandatory; or that operational uncertainty can be eliminated.

Specifically, the paper does not claim that Article 41 imposes a positive obligation to equip a system with recognition capability. Nor does it claim that the recognition window is content of the prohibition rather than a precondition of its practical operability.

The argument is determinate but bounded. It establishes that existing IHL obligations — the prohibition on attacking persons hors de combat, customary and reflected in Rule 47 and Article 23(c) of the Hague Regulations; the treaty formulation of Article 41(1) and verification of Article 57(2)(a)(i) for States party; and legal review where applicable — remain operative in AI-mediated operational environments; that the temporal

compression of decision cycles can render human oversight nominal rather than substantive; that certain safeguard categories serve compliance-relevant operational functions; and that the systematic exclusion of surrender-sensitive safeguards from systems where they are technically achievable — assessed against verifiable maturity and operational criteria — engages those duties and requires documented justification. The paper neither advocates categorical AI restrictions nor accepts the proposition that fewer safeguards are equivalent to greater operational capability or legal safety. The translation of IHL principles into technical design specifications raises complex questions that remain unresolved in both legal scholarship and technical practice; the argument presented here is offered as a contribution to that discussion, not as a resolution of it.

KEY TAKEAWAYS FOR PROCUREMENT AND COMMAND STAFF

1. Source — The prohibition on attacking persons hors de combat is customary international law (Rule 47; Art. 23(c) Hague 1907). It binds all parties, irrespective of treaty ratification.
2. Recognition Window — The prohibition operates upon recognition. Recognition requires a temporal interval. Where system design compresses the engagement below that interval, the practical operability of the rule is foreclosed.
3. Treaty Formulation — For States party to AP I, Art. 41 provides the clearest formulation; Art. 57 supplies precautionary support; Art. 36 provides the review procedure. For other States, the question arises under national procedures (US DoD Directive 3000.09, UK Ambitious, Safe, Responsible, national procurement standards).
4. Design Choice — An impossibility established at the design stage is not a circumstance encountered in the field. It is a decision taken, and a decision can be examined.
5. Review — Weapons review (Art. 36 for States party, or national equivalent) must determine whether the system preserves the recognition window in foreseeable conditions of use.
6. Safeguards — Surrender-recognition capability is not an ethical add-on but the effectiveness condition of obligations that have bound parties for over a century. Its systematic exclusion engages existing duties and warrants documented justification.

X. CONCLUSION

The framing that fewer safeguards necessarily increase operational effectiveness or national security is incomplete. Certain safeguards function as operational mechanisms

through which existing humanitarian obligations remain capable of being discharged — and their absence is a fact the law already has the instruments to examine.

The compliance challenges associated with autonomous systems are not solely questions of accuracy or capability. They are questions of time. Human judgment operates within biological limits, while computational systems increasingly operate at machine speed. Where this temporal asymmetry becomes sufficiently pronounced, meaningful human control becomes nominal rather than substantive, and compliance-relevant functions must be preserved within the architecture, because architectural safeguards may be among the few mechanisms operating within the window in which the decision is actually made.

Surrender recognition capability in operational AI systems is the paradigm case of such a lawful operational safeguard. The law does not need to be rewritten. Its existing protections must remain executable. No one can be required to do the impossible — *ad impossibilia nemo tenetur* — but an impossibility established at the design stage is not a circumstance encountered. It is a decision taken, and a decision can be examined.

Preserving a recognition window is one concrete way of ensuring that autonomous functions do not deprive a person of the practical possibility of surrendering, do not deprive the rule of distinction of its elementary operation, and do not deprive the prohibition binding on every party to an armed conflict of its intended effect.

If the analysis presented here is correct, the relevant question in future governance, procurement, and accountability discussions is not whether AI systems should have fewer or more safeguards as a general matter. It is which safeguards perform compliance-relevant functions under existing law, what verifiable criteria determine their technical feasibility, and how the sequence technical feasibility → documented justification → command awareness → operational accountability structures legal review under existing obligations and, where applicable, formal legal-review duties.

METHODOLOGICAL NOTE

This paper proposes an analytical framework grounded in treaty text; it does not purport to resolve disputed questions of treaty interpretation, nor to establish binding standards of conduct beyond those the cited provisions themselves impose. The qualifications that earlier editions distributed throughout the text are consolidated here: the paper advances no claim that any specific technical implementation is legally mandated; the illustrative scenarios in Section V.D are analytical constructs and do not describe or evaluate any existing system or deployment; the temporal figures in Section IV are drawn from published literature and are presented as orders of magnitude establishing a structural asymmetry, not as operational specifications; references to procurement frameworks and audit criteria are analytical reference points, not legal advice; and the analysis rests on customary international law, reflected in Rule 47 and Article 23(c) of

the Hague Regulations, with Article 41 providing the clearest treaty formulation for States party to Additional Protocol I and Article 36 providing the procedural anchor where applicable; comparable inquiries for non-party States rest on separate, expressly identified legal bases. The author welcomes engagement, critique, and further development of the framework by scholars, practitioners, and policymakers working in international humanitarian law, autonomous systems governance, and defence procurement.

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Giovanni Nardacci is the founder of Human Flag Association, a Switzerland-based humanitarian technical initiative focused on civilian protection standards and IHL compliance architectures in AI-enabled operational environments. UNGM Partner No. 6126. humanflag.org. The author prepared this paper independently; no funding was received from any party with a commercial or governmental interest in the subject matter.

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